Electronic Gaps Near Quasi-Degeneracy: Minimal H_2 Diagnostic, Ethylene Exploration in inZOR-ND, and the Discrete-Search Benchmark Boundary

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Abstract

Near **quasi-degeneracy**, multi-reference methods depend critically on the **active space**. Poor orbital choices can distort **excitation gaps** and state orderings long before a full potential-energy surface is trusted. This note unifies two complementary **inZOR-ND** threads under the same chemical motif—gaps under electronic stress—while keeping their **algorithms and claims** distinct.

First, we document a **minimal H_2 diagnostic**: a seven-point H–H stretch (cc-pVDZ) with **SA-CASSCF(2,2)** HOMO–LUMO gaps comparing active MO indices chosen by **inZOR-ND** to a **NOON**-style rule. The inZOR-ND track yields a **monotone** gap curve across the scan; the NOON-tracked pair does **not**, with a documented spike at $R = 2.0$ Å ($\sim$18.14 eV) versus $\sim$2 eV at neighbouring points.

Second, we summarise **ethylene** near torsional quasi-degeneracy in the **continuous exploration** stack: SA-CASSCF(2,2) gap-style observables feed an evolving population (2D/3D, multiple seeds, score variants) in the dedicated scene `scene_ethylene_quasidegen`. We reference exported validation figures and a separate **3D regional** technical report on multi-basin low-gap structure, discovery speed versus uniform control, and reproducibility limits for identical probe traces.

We state **claim hygiene**: continuous population logs must not be cited as **discrete orbital-set search** benchmark outcomes. Large-scale discrete comparisons (e.g. eight-benchmark suite including ethylene SA-CASSCF and N_2) remain the citation target for CAS discovery claims [1].

All tabulated H_2 numbers are frozen in machine-readable JSON alongside the public HTML companion [2]; the 3D ethylene regional study is published as a sibling page [3].

1 Motivation

Multi-reference models are sensitive to orbital partitioning: the same nominal method can produce qualitatively wrong gaps along a dissociation coordinate if the active pair is misidentified [4]. Practical workflows therefore need **cheap diagnostics** that stress-test gap behaviour before expensive dynamics or extended surfaces.

inZOR-ND is used in this work in two senses that share a scientific theme (gaps near quasi-degeneracy) but differ in implementation:

- **Discrete search** over candidate active spaces (orbital index sets), as in published CAS benchmarks [1].

- **Continuous exploration**: internal coordinates evolve under selection pressure with quantum-chemistry probes scoring each proposal (ethylene scene).

Confusing the two would mis-state what was validated where; we keep the boundary explicit throughout.

2 Methods

2.1 H_2: seven-point SA-CASSCF(2,2) gap scan

Hydrogen in a minimal (2e, 2o) active space isolates **orbital-choice effects** on a single gap observable without conformational degrees of freedom. We use PySCF-based SA-CASSCF(2,2) [5] in the cc-pVDZ basis at bond distances $\{0.7, 0.9, 1.1, 1.3, 1.6, 2.0, 2.5\}$ Å. Two orbital choices are compared:

- **inZOR-ND track:** active MO indices $(0, 1)$.
- **NOON-style track:** indices $(0, 3)$ for this documented run.

HOMO–LUMO gaps (eV) and monotonicity flags are archived in `h2_stretch_results.json` [2].

2.2 Ethylene: continuous exploration scene

Ethylene near twisted π quasi-degeneracy is a standard setting where single-reference pictures weaken; SA-CASSCF(2,2) provides a minimal multi-reference anchor. In **inZOR-ND**, genomes propose moves in internal coordinates; a quantum-chemistry probe evaluates gap-style scores (e.g. S_0–S_1) that feed fitness. Engineering details (score variants `legacy`, `local_coherent`, `dwell_coherent`; optional 3D `ZOR_ETHYLENE_ND`) and reproduction paths are documented on the public report [2]. The companion **3D regional** study pools low-gap probes for DBSCAN, reports per-seed tables, and compares discovery speed at gap < 10 meV to uniform random sampling on a matched budget [3].

3 Results

3.1 H_2 gaps and monotonicity

Table 1 reproduces the verified JSON entries (rounded for display). The inZOR-ND track is **monotone decreasing** along dissociation; the NOON track is **not**, with the pathology at $R = 2.0$ Å.

Table 1: SA-CASSCF(2,2) HOMO–LUMO gap (eV) along the H–H scan (cc-pVDZ). Source: `h2_stretch_results.json` [2].

R (Å)	inZOR gap (eV)	NOON gap (eV)
0.7	10.743743	10.743759
0.9	7.974340	7.974299
1.1	5.677354	5.677354
1.3	3.887806	3.887805
1.6	2.049057	2.049044
2.0	0.764453	**18.143451**
2.5	0.188814	0.188821

Remark. The NOON value at 2.0 Å is an **early warning**: neighbouring points lie near $\sim$2 eV, so the spike signals inconsistent active-pair tracking rather than a trustworthy local gap for that

geometry.

3.2 Ethylene: exported validation visuals

Figures 1–3 are taken from the same validation thread as the public unified report [2]. They illustrate gap-driven structure in the ethylene workflow; quantitative multi-seed regional tables, clustering, and random-baseline comparisons are consolidated in the 3D technical companion [3].

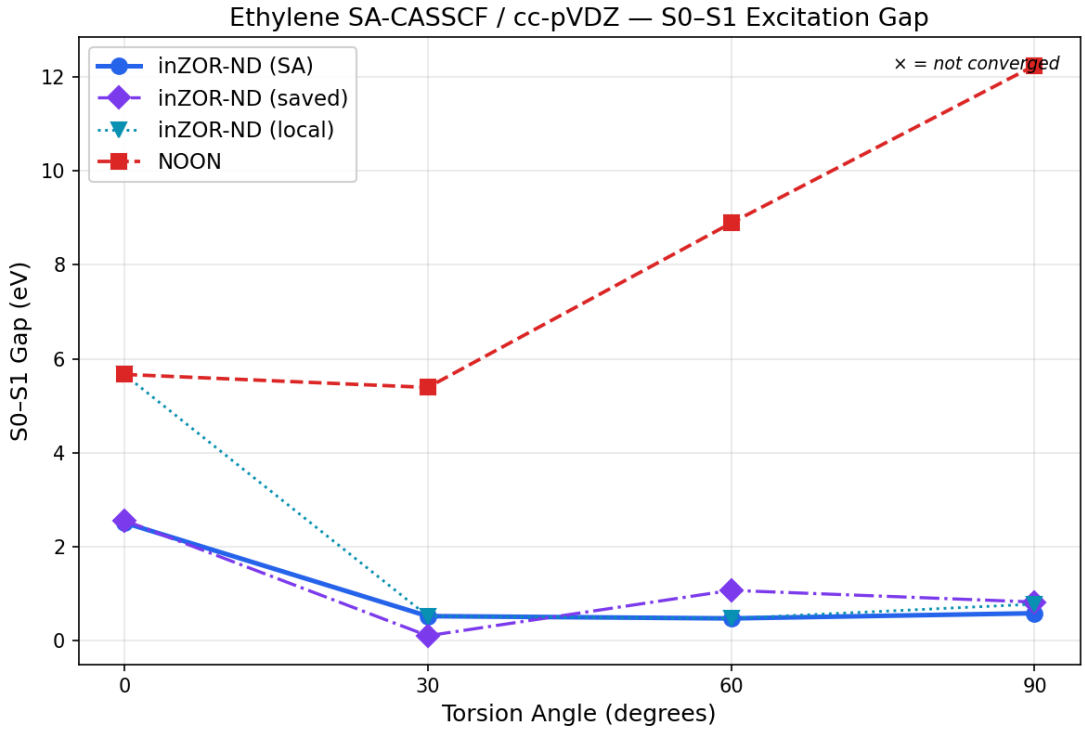

Figure 1: S_0–S_1 gap visual (ethylene quasi-degeneracy study, exported from the inZOR-ND validation workspace).

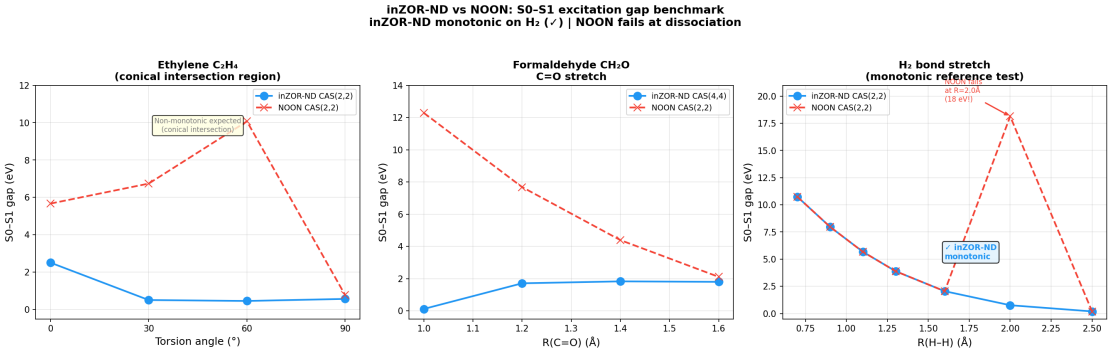

Figure 2: Validation-style comparison across batches or score variants (ethylene thread export).

4 Relation to discrete-search benchmarks

The **eight-benchmark** interactive report treats ethylene SA-CASSCF within a **discrete-search** protocol against NOON-MP2 / AVAS-style baselines [1]. The present unified chapter adds (i) a **minimal H_2 gap sanity check** for orbital heuristics and (ii) a **continuous exploration** narrative for ethylene. Readers should cite the comparative suite for large-molecule

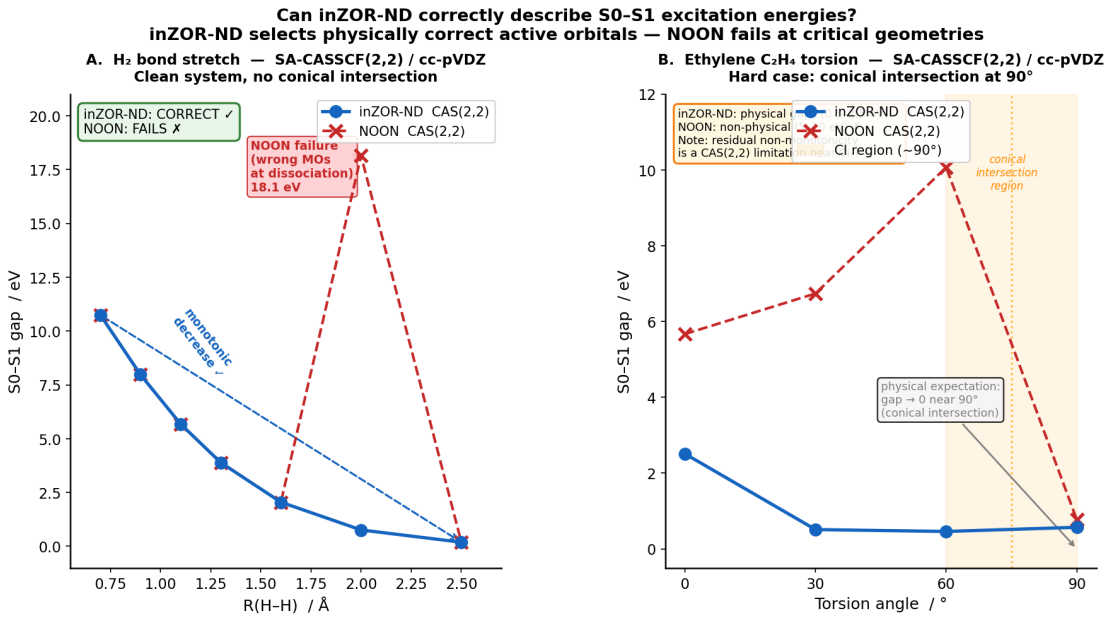

Figure 3: Companion metrics / reference panel for the same validation workflow.

CAS discovery statistics, and this note for gap-focused motivation, the H_2 table, and the exploration/validation figures.

5 Reproducibility

H_2: regenerate from the driver that produced `h2_stretch_results.json`; pin PySCF and dependencies for bit-identical replication if required.

Ethylene: scene assets, exports, and validation logs live under `zor_scenes/scene_ethylene_quasidegen/` in the source repository; see the public HTML companion [2].

3D regional batch: named validation directories and JSON summaries are listed on the 3D page [3]. Reproducibility of clustering is demonstrated for **byte-identical** `probes.jsonl` reruns, not for independent stochastic trajectories.

6 Conclusion

Quasi-degenerate regimes demand scrutiny of **gap observables**, not only total energies. A seven-point H_2 scan separates inZOR-ND and NOON-style orbital choices on a transparent curve: one track stays monotone; the other breaks down at an intermediate bond length. In parallel, ethylene provides a richer geometry where inZOR-ND continuous exploration couples SA-CASSCF(2,2) gaps to population dynamics, with extended regional analysis in a dedicated 3D report. Keeping **discrete-search** and **continuous-exploration** claims apart preserves scientific hygiene as both lines evolve.

Data and code availability

Interactive report (HTML):
https://dumitrunovic-svg.github.io/inZOR-ND/tests/qc_gaps_h2_ethylene_unified/

Ethylene 3D technical page:
https://dumitrunovic-svg.github.io/inZOR-ND/tests/qc_gaps_h2_ethylene_unified/ethylene_3d_zor.html

Repository (tests and brochure source):
https://github.com/dumitrunovic-svg/inZOR-ND.

References

[1] Dumitru Novic. inZOR-ND: comparative validation for automatic CASSCF active space selection. Interactive benchmark report (GitHub Pages), 2026. https://dumitrunovic-svg.github.io/inZOR-ND/tests/active_space_comparative/.

[2] Dumitru Novic. QC gaps: minimal H_2 diagnostic, ethylene exploration in inZOR-ND, and discrete-search benchmarks. Interactive report (GitHub Pages), 2026. https://dumitrunovic-svg.github.io/inZOR-ND/tests/qc_gaps_h2_ethylene_unified/.

[3] Dumitru Novic. Ethylene (3D) — quasi-degenerate gap structure via inZOR-ND. Technical report (GitHub Pages), 2026. https://dumitrunovic-svg.github.io/inZOR-ND/tests/qc_gaps_h2_ethylene_unified/ethylene_3d_zor.html.

[4] Trygve Helgaker, Poul Jørgensen, and Jeppe Olsen. *Molecular Electronic-Structure Theory*. John Wiley & Sons, 2000.

[5] Qiming Sun et al. PySCF: the python-based simulations of chemistry framework. *WIREs Computational Molecular Science*, 8(1):e1340, 2018. doi: 10.1002/wcms.1340.